

Observation Task 4: Student Result and Performance Analyzer using Java (CO1)

Title

Student Performance Analysis System

Objective / Aim

To design and implement a student performance analysis system in Java using Object-Oriented Programming principles to calculate marks, percentage, grade, highest and lowest marks, and overall academic performance.

Problem Statement

Develop a Java application for a college to analyze the academic performance of a student.

The system should:

- Store subject marks using an array.
- Calculate total and average marks.
- Find the highest and lowest marks.
- Calculate percentage.
- Assign a grade based on percentage.
- Determine whether the student has passed or failed.
- Display the student's name in a formatted manner using String operations.
- Round the percentage to two decimal places using the `Math` package.
- Display a performance remark based on the student's percentage.
- Use a class to represent the student and demonstrate basic Object-Oriented Programming.

Concepts Used

Concept	Description
Class	<code>Student</code> class represents student information
Object	<code>Student</code> object stores student details
Constructor	Initializes roll number, name and marks

Concept	Description
Array	Stores marks of five subjects
Methods	Perform calculations and display results
String Handling	<code>trim()</code> , <code>toUpperCase()</code> , <code>length()</code>
Math Methods	<code>Math.max()</code> , <code>Math.min()</code> , <code>Math.round()</code>
Loop	Calculates total and displays subject marks
Conditional Statements	Determines grade, result and remark

Data Members

The `Student` class contains:

```
rollNumber
studentName
marks[]
```

Grade Criteria

Percentage Grade Result

90–100	A+	PASS
80–89	A	PASS
70–79	B	PASS
60–69	C	PASS
50–59	D	PASS
Below 50	F	FAIL

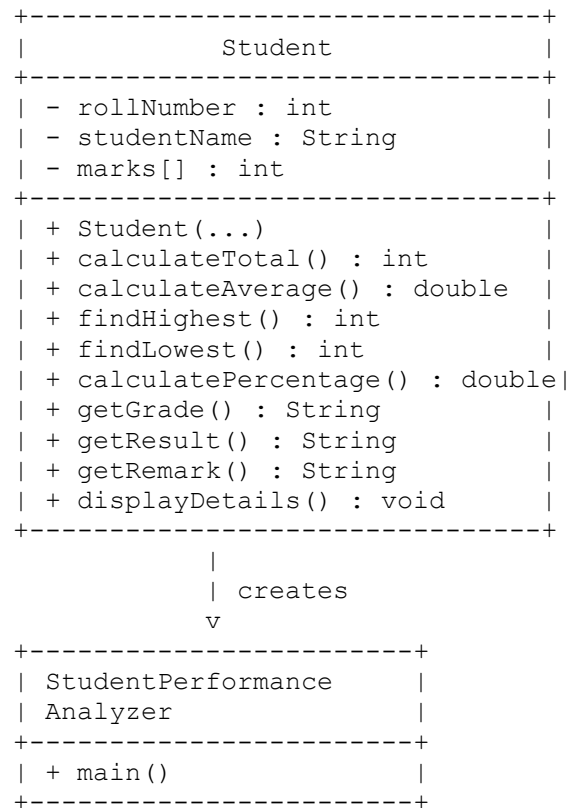
A student is considered **PASS** when the percentage is at least 50%.

Algorithm / Logic Description

1. Create a class named `Student`.
2. Declare `rollNumber`, `studentName`, and an integer array `marks`.

3. Create a constructor to initialize the student details.
4. Create a `calculateTotal()` method to calculate the total marks using a loop.
5. Create a `calculateAverage()` method to calculate the average.
6. Create a `findHighest()` method using `Math.max()`.
7. Create a `findLowest()` method using `Math.min()`.
8. Calculate the percentage from the total marks.
9. Use `Math.round()` to round the percentage to two decimal places.
10. Use conditional statements to determine the grade.
11. Determine PASS or FAIL based on a 50% threshold.
12. Generate a performance remark based on the percentage.
13. Use `trim()`, `toUpperCase()`, and `length()` to format the student's name.
14. Create a `Student` object in the `main()` method.
15. Display the complete student performance report.

UML / Class Diagram



Source Code

```

package tasks;

class Student {
    int rollNumber;
    String studentName;
    int[] marks;
  
```

```

Student(int rollNumber, String studentName, int[] marks) {
    this.rollNumber = rollNumber;
    this.studentName = studentName;
    this.marks = marks;
}

int calculateTotal() {
    int total = 0;

    for (int mark : marks) {
        total = total + mark;
    }

    return total;
}

double calculateAverage() {
    return (double) calculateTotal() / marks.length;
}

int findHighest() {
    int highest = marks[0];

    for (int mark : marks) {
        highest = Math.max(highest, mark);
    }

    return highest;
}

int findLowest() {
    int lowest = marks[0];

    for (int mark : marks) {
        lowest = Math.min(lowest, mark);
    }

    return lowest;
}

double calculatePercentage() {
    double percentage =
        ((double) calculateTotal() / (marks.length * 100)) * 100;

    return Math.round(percentage * 100.0) / 100.0;
}

String getGrade() {
    double percentage = calculatePercentage();

    if (percentage >= 90) {
        return "A+";
    } else if (percentage >= 80) {
        return "A";
    } else if (percentage >= 70) {
        return "B";
    }
}

```

```

        } else if (percentage >= 60) {
            return "C";
        } else if (percentage >= 50) {
            return "D";
        } else {
            return "F";
        }
    }

String getResult() {
    if (calculatePercentage() >= 50) {
        return "PASS";
    } else {
        return "FAIL";
    }
}

String getRemark() {
    double percentage = calculatePercentage();

    if (percentage >= 90) {
        return "Excellent Performance";
    } else if (percentage >= 80) {
        return "Very Good Performance";
    } else if (percentage >= 70) {
        return "Good Performance";
    } else if (percentage >= 60) {
        return "Satisfactory Performance";
    } else if (percentage >= 50) {
        return "Needs Improvement";
    } else {
        return "Poor Performance";
    }
}

void displayDetails() {
    String formattedName = studentName.trim().toUpperCase();

    System.out.println("===== STUDENT PERFORMANCE REPORT =====");
    System.out.println("Roll Number      : " + rollNumber);
    System.out.println("Student Name      : " + formattedName);
    System.out.println("Name Length       : " + formattedName.length());

    System.out.println("\nSubject Marks:");

    for (int i = 0; i < marks.length; i++) {
        System.out.println("Subject " + (i + 1) + "      : " +
marks[i]);
    }

    System.out.println("\nTotal Marks      : " + calculateTotal());
    System.out.println("Average Marks    : " + calculateAverage());
    System.out.println("Highest Marks    : " + findHighest());
    System.out.println("Lowest Marks     : " + findLowest());
    System.out.println("Percentage       : " + calculatePercentage() +
"%");

    System.out.println("Grade           : " + getGrade());
}

```

```

        System.out.println("Result          : " + getResult());
        System.out.println("Remark         : " + getRemark());
    }
}

public class StudentPerformanceAnalyzer {
    public static void main(String[] args) {

        int[] marks = {85, 92, 78, 88, 95};

        Student student = new Student(
            101,
            "Rahul Sharma ",
            marks
        );

        student.displayDetails();
    }
}

```

Compilation and Execution

Since the program contains:

```
package tasks;
```

the file should be saved as:

```
tasks/StudentPerformanceAnalyzer.java
```

Compile

```
javac tasks/StudentPerformanceAnalyzer.java
```

Run

```
java tasks.StudentPerformanceAnalyzer
```

Sample Output

```

===== STUDENT PERFORMANCE REPORT =====
Roll Number      : 101
Student Name     : RAHUL SHARMA
Name Length      : 12

Subject Marks:
Subject 1        : 85
Subject 2        : 92
Subject 3        : 78
Subject 4        : 88
Subject 5        : 95

Total Marks      : 438
Average Marks    : 87.6
Highest Marks    : 95

```

Lowest Marks : 78
Percentage : 87.6%
Grade : A
Result : PASS
Remark : Very Good Performance

Calculation

Total Marks

$$85 + 92 + 78 + 88 + 95 = 438$$

Average

$$438 / 5 = 87.6$$

Percentage

Since there are five subjects and each subject has a maximum of 100 marks:

$$\begin{aligned}\text{Percentage} &= (438 / 500) \times 100 \\ &= 87.6\%\end{aligned}$$

Highest and Lowest

$$\begin{aligned}\text{Highest} &= 95 \\ \text{Lowest} &= 78\end{aligned}$$

Grade

Since the percentage is **87.6%**, the grade is:

A

Result

Since the percentage is greater than or equal to 50%:

PASS

Performance Remark

For a percentage between 80% and 89%:

Very Good Performance

Real-Time Application

This type of Student Performance Analysis System can be used in:

- Colleges and universities
- Schools
- Examination management systems
- Student information systems
- Academic performance tracking applications

Result

The **Student Performance Analysis System** was successfully implemented in Java using Object-Oriented Programming concepts, arrays, String methods, and built-in `Math` methods.

The program successfully calculates and displays the student's **total marks, average, highest marks, lowest marks, percentage, grade, result, and performance remark**.

Conclusion

Thus, the Java program successfully demonstrates the use of **classes, objects, constructors, arrays, methods, loops, conditional statements, String handling, and Math methods** for analyzing student academic performance.